

VK-RCD ??? HPE ? HAR ?????

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2026 年 5 月 7 日

摘要

???? VK-RCD ??? HPE ? HAR ?????????????? X-Fi ????????HPE ??????????????????????
X-Fi ?????? ????? X-Fi ??????HPE ?? MPJPE ? PA-MPJPE????????HAR ?? Accu-
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??? DONE ?????????????????????????????? MISSING_MODEL ? MISSING_EVAL
?????????????????????

表 1: HPE ? HAR ????????

??	DONE	MISSING_MODEL	MISSING_EVAL
HPE	8	18	20
HAR	24	1	0

表 2: ???????

??	??	??	??	??
HPE	model	ablation/no_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\no_kd
HPE	model	ablation/output_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\output_kd
HPE	model	ablation/output_token_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\output_token_kd
HPE	model	ablation/output_bone_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\output_bone_kd
HPE	model	ablation/full_structural_kd/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\full_structural_kd
HPE	model	ablation/uniform/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\uniform
HPE	model	ablation/attention/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\attention
HPE	model	ablation/uncertainty/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\uncertainty
HPE	model	ablation/mlp_vk/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\mlp_vk
HPE	model	ablation/skeleton_prompt/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\ablation\skeleton_prompt
HPE	model	teacher_full_cross_scene_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\teacher_full_cross_scene_split
HPE	model	student_vk_missing_cross_scene_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_vk_missing_cross_scene_split
HPE	model	student_nv_missing_cross_scene_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_nv_missing_cross_scene_split
HPE	model	baseline_full_cross_scene_split/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\baseline_full_cross_scene_split
HPE	model	teacher_full_cross_subject/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\teacher_full_cross_subject
HPE	model	student_vk_missing_cross_subject/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_vk_missing_cross_subject
HPE	model	student_nv_missing_cross_subject/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\student_nv_missing_cross_subject
HPE	model	baseline_full_cross_subject/best.pth	MISSING_MODEL	E:\Deskbook\Tea\HPE\outputs\baseline_full_cross_subject
HPE	eval	student_nv_nonvisual_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\student_nv_nonvisual_combinations.csv
HPE	eval	baseline_full_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\baseline_full_all_combinations.csv
HPE	eval	ablation_distillation_no_kd_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_distillation_no_kd_combinations.csv
HPE	eval	ablation_distillation_output_kd_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_distillation_output_kd_combinations.csv
HPE	eval	ablation_distillation_output_token_kd_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_distillation_output_token_kd_combinations.csv
HPE	eval	ablation_distillation_output_bone_kd_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_distillation_output_bone_kd_combinations.csv
HPE	eval	ablation_distillation_full_structural_kd_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_distillation_full_structural_kd_combinations.csv
HPE	eval	ablation_reliability_uniform_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_reliability_uniform_combinations.csv
HPE	eval	ablation_reliability_attention_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_reliability_attention_combinations.csv
HPE	eval	ablation_reliability_uncertainty_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_reliability_uncertainty_combinations.csv
HPE	eval	ablation_encoder_mlp_vk_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_encoder_mlp_vk_combinations.csv
HPE	eval	ablation_encoder_skeleton_prompt_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\ablation_encoder_skeleton_prompt_combinations.csv
HPE	eval	teacher_cross_scene_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\teacher_cross_scene_all_combinations.csv
HPE	eval	student_vk_cross_scene_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\student_vk_cross_scene_all_combinations.csv
HPE	eval	student_nv_cross_scene_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\student_nv_cross_scene_all_combinations.csv
HPE	eval	baseline_cross_scene_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\baseline_cross_scene_all_combinations.csv
HPE	eval	teacher_cross_subject_all_combinations.csv	MISSING_EVAL	E:\Deskbook\Tea\HPE\outputs\eval\teacher_cross_subject_all_combinations.csv

??	??	??	??	??
HPE	eval	student_vk_cross_subject_missing_evaluations.csv	MISSING EVALUATIONS	E:\Deskbook\Tea\HPE\outputs\eval\student_vk_cross_subject_missing_evaluations.csv
HPE	eval	student_nv_cross_subject_missing_evaluations.csv	MISSING EVALUATIONS	E:\Deskbook\Tea\HPE\outputs\eval\student_nv_cross_subject_missing_evaluations.csv
HPE	eval	baseline_cross_subject_all_missing_evaluations.csv	MISSING EVALUATIONS	E:\Deskbook\Tea\HPE\outputs\eval\baseline_cross_subject_all_missing_evaluations.csv
HAR	model	student_vk_missing_cross_subject_test_model	MISSING MODEL	E:\Deskbook\Tea\HAR\outputs\student_vk_missing_cross_subject_test_model

2 HPE ????

?? HPE ???????? 62 ???????? MPJPE ??? X-Fi ??? 34 ??PA-MPJPE ??? X-Fi
 ??? 33 ??HPE ????????????????

表 3: HPE ???????? X-Fi ??

??	??	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ
VK-RCD-Teacher	teacher_all_combinations.csv	48.45	83.70	-35.25	
VK-RCD-Student-VK	all_combinations.csv	50.17	83.70	-33.53	

表 4: HPE ??? X-Fi ????????

??	??	????	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ
VK-RCD-Teacher	random_splitvk		487.99	93.90	+394.09	+7
VK-RCD-Teacher	random_splitdepth		234.48	101.80	+132.68	+3
VK-RCD-Teacher	random_splitlidar		1615.42	167.10	+1448.32	+28
VK-RCD-Teacher	random_splitmmwave		4196.64	127.40	+4069.24	+22
VK-RCD-Teacher	random_splitwifi-csi		4173.33	225.60	+3947.73	+27
VK-RCD-Teacher	random_splitvk+depth		369.12	86.10	+283.02	+4
VK-RCD-Teacher	random_splitvk+lidar		1072.74	93.00	+979.74	+15
VK-RCD-Teacher	random_splitvk+mmwave		814.71	88.80	+725.91	+11
VK-RCD-Teacher	random_splitvk+wifi-csi		2491.83	93.00	+2398.83	+27
VK-RCD-Teacher	random_splitdepth+lidar		285.80	102.50	+183.30	+3
VK-RCD-Teacher	random_splitdepth+mmwave		178.87	98.00	+80.87	+1
VK-RCD-Teacher	random_splitdepth+wifi-csi		197.61	101.80	+95.81	+2
VK-RCD-Teacher	random_splitlidar+mmwave		2157.14	109.80	+2047.34	+32
VK-RCD-Teacher	random_splitlidar+wifi-csi		4007.08	159.50	+3847.58	+28
VK-RCD-Teacher	random_splitmmwave+wifi-csi		4675.91	117.20	+4558.71	+21
VK-RCD-Teacher	random_splitvk+depth+lidar		634.60	84.80	+549.80	+6

??	??	????	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ
VK-RCD-Teacher	random_splitvk+depth+mmwave		465.16	83.40	+381.76	+3
VK-RCD-Teacher	random_splitvk+depth+wifi- csi		100.71	85.30	+15.41	+
VK-RCD-Teacher	random_splitvk+lidar+mmwave		1035.05	88.40	+946.65	+18
VK-RCD-Teacher	random_splitvk+lidar+wifi- csi		4736.11	93.00	+4643.11	+21
VK-RCD-Teacher	random_splitvk+mmwave+wifi- csi		4769.71	88.50	+4681.21	+19
VK-RCD-Teacher	random_splitdepth+lidar+mmwave		310.49	96.00	+214.49	+2
VK-RCD-Teacher	random_splitdepth+lidar+wifi- csi		177.40	102.00	+75.40	+1
VK-RCD-Teacher	random_splitdepth+mmwave+wifi- csi		121.88	97.00	+24.88	+
VK-RCD-Teacher	random_splitlidar+mmwave+wifi- csi		4360.45	107.40	+4253.05	+25
VK-RCD-Teacher	random_splitvk+depth+lidar+mmwave		593.34	83.50	+509.84	+4
VK-RCD-Teacher	random_splitvk+depth+lidar+wifi- csi		67.89	86.00	-18.11	-
VK-RCD-Teacher	random_splitvk+depth+mmwave+wifi- csi		71.27	84.00	-12.73	-
VK-RCD-Teacher	random_splitvk+lidar+mmwave+wifi- csi		4876.08	88.60	+4787.48	+18
VK-RCD-Teacher	random_splitdepth+lidar+mmwave+wifi- csi		112.99	97.60	+15.39	+
VK-RCD-Teacher	random_splitvk+depth+lidar+mmwave+ csi		48.45	83.70	-35.25	-1
VK-RCD-Student-VK	random_splitvk		92.66	93.90	-1.24	-1
VK-RCD-Student-VK	random_splitdepth		53.88	101.80	-47.92	-1
VK-RCD-Student-VK	random_splitlidar		139.90	167.10	-27.20	-
VK-RCD-Student-VK	random_splitmmwave		114.98	127.40	-12.42	-1
VK-RCD-Student-VK	random_splitwifi-csi		222.14	225.60	-3.46	+

??	??	????	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ
VK-RCD-Student-VK	random_splitvk+depth		51.65	86.10	-34.45	-1
VK-RCD-Student-VK	random_splitvk+lidar		73.18	93.00	-19.82	-1
VK-RCD-Student-VK	random_splitvk+mmwave		68.93	88.80	-19.87	-1
VK-RCD-Student-VK	random_splitvk+wifi-csi		88.11	93.00	-4.89	-1
VK-RCD-Student-VK	random_splitdepth+lidar		53.00	102.50	-49.50	-1
VK-RCD-Student-VK	random_splitdepth+mmwave		52.32	98.00	-45.68	-1
VK-RCD-Student-VK	random_splitdepth+wifi-csi		53.54	101.80	-48.26	-1
VK-RCD-Student-VK	random_splitlidar+mmwave		97.54	109.80	-12.26	-1
VK-RCD-Student-VK	random_splitlidar+wifi-csi		134.11	159.50	-25.39	-1
VK-RCD-Student-VK	random_splitmmwave+wifi-csi		108.54	117.20	-8.66	-1
VK-RCD-Student-VK	random_splitvk+depth+lidar		51.10	84.80	-33.70	-1
VK-RCD-Student-VK	random_splitvk+depth+mmwave		50.50	83.40	-32.90	-1
VK-RCD-Student-VK	random_splitvk+depth+wifi-csi		51.63	85.30	-33.67	-1
VK-RCD-Student-VK	random_splitvk+lidar+mmwave		66.08	88.40	-22.32	-1
VK-RCD-Student-VK	random_splitvk+lidar+wifi-csi		70.68	93.00	-22.32	-1
VK-RCD-Student-VK	random_splitvk+mmwave+wifi-csi		66.69	88.50	-21.81	-1
VK-RCD-Student-VK	random_splitdepth+lidar+mmwave		51.92	96.00	-44.08	-1
VK-RCD-Student-VK	random_splitdepth+lidar+wifi-csi		52.86	102.00	-49.14	-1

??	??	????	Ours MPJPE	X-Fi MPJPE	Δ MPJPE	Δ
VK-RCD-Student-VK	random_splitdepth+mmwave+wifi-csi		52.13	97.00	-44.87	-1
VK-RCD-Student-VK	random_splitlidar+mmwave+wifi-csi		94.20	107.40	-13.20	-1
VK-RCD-Student-VK	random_splitvk+depth+lidar+mmwave		50.25	83.50	-33.25	-1
VK-RCD-Student-VK	random_splitvk+depth+lidar+wifi-csi		51.05	86.00	-34.95	-1
VK-RCD-Student-VK	random_splitvk+depth+mmwave+wifi-csi		50.41	84.00	-33.59	-1
VK-RCD-Student-VK	random_splitvk+lidar+mmwave+wifi-csi		64.22	88.60	-24.38	-1
VK-RCD-Student-VK	random_splitdepth+lidar+mmwave+wifi-csi		51.78	97.60	-45.82	-1
VK-RCD-Student-VK	random_splitvk+depth+lidar+mmwave+wifi-csi		50.17	83.70	-33.53	-1

2.1 HPE VK ???????????

表 5: Student-VK ? VK ???????????????

??	Noise Std	Drop Prob	MPJPE(mm)	PA-MPJPE(mm)	M
student_vk_noise_100ustness.csv			50.18	33.48	0.0014590788163282
student_vk_noise_100ustness.csv			52.64	34.57	0.0015710848316534
student_vk_noise_100ustness.csv			54.10	35.19	0.0016379341990311
student_vk_noise_100ustness.csv			54.75	35.50	0.0016690452336497
student_vk_noise_50ustness.csv			50.88	33.92	0.001488102523395
student_vk_noise_50ustness.csv			52.88	34.71	0.001581960488836
student_vk_noise_50ustness.csv			54.12	35.19	0.0016401267167432
student_vk_noise_50ustness.csv			54.76	35.50	0.0016697228733739
student_vk_noise_100ustness.csv			52.08	34.57	0.001539294305564
student_vk_noise_100ustness.csv			53.26	34.93	0.0015987949889641
student_vk_noise_100ustness.csv			54.12	35.20	0.001638386860237
student_vk_noise_100ustness.csv			54.74	35.52	0.0016693329612529
student_vk_noise_200ustness.csv			53.63	35.20	0.001611003164998
student_vk_noise_200ustness.csv			53.85	35.15	0.001621275579426

??	Noise Std	Drop Prob	MPJPE(mm)	PA-MPJPE(mm)	M
student_vk_noise_r200.0	stness.csv		54.21	35.25	0.0016435792571283
student_vk_noise_r200.0	stness.csv		54.78	35.54	0.0016701842608368

3 HAR ????

?? HAR ????????? 75 ????????? Accuracy ???? X-Fi ???? 50 ??HAR ?????????Accuracy
?????

表 6: HAR ????????? X-Fi ??

??	??	Ours Acc.
VK-RCD-HAR-Teacher	teacher_random_all_combinations.csv	95.57
VK-RCD-HAR-Student-VK-Missing	student_vk_random_all_combinations.csv	96.56
VK-RCD-HAR-Teacher	all_combinations.csv	95.59
VK-RCD-HAR-Ablation-NoKD	ablation_no_distill_all_combinations.csv	95.99
VK-RCD-HAR-Ablation-UniformFusion	ablation_uniform_fusion_all_combinations.csv	95.39

表 7: HAR ??? X-Fi ?????????

??	??	????	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD-HAR-Teacher	random_splitvk		3.40	26.50	-23.10	0.45
VK-RCD-HAR-Teacher	random_splitdepth		83.71	48.10	+35.61	82.80
VK-RCD-HAR-Teacher	random_splitlidar		3.36	52.70	-49.34	1.17
VK-RCD-HAR-Teacher	random_splitmmwave		66.72	85.70	-18.98	64.78
VK-RCD-HAR-Teacher	random_splitvk+depth		83.50	45.30	+38.20	82.60
VK-RCD-HAR-Teacher	random_splitvk+lidar		3.91	35.20	-31.29	1.31
VK-RCD-HAR-Teacher	random_splitvk+mmwave		66.46	73.40	-6.94	63.67
VK-RCD-HAR-Teacher	random_splitdepth+lidar		83.37	51.60	+31.77	82.54

??	??	????	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD-HAR-Teacher	random_splitdepth+mmwave		95.76	79.80	+15.96	95.74
VK-RCD-HAR-Teacher	random_splitlidar+mmwave		66.69	88.70	-22.01	63.96
VK-RCD-HAR-Teacher	random_splitvk+depth+lidar		83.66	48.70	+34.96	82.91
VK-RCD-HAR-Teacher	random_splitvk+depth+mmwave		95.66	70.70	+24.96	95.65
VK-RCD-HAR-Teacher	random_splitvk+lidar+mmwave		66.39	77.80	-11.41	63.36
VK-RCD-HAR-Teacher	random_splitdepth+lidar+mmwave		95.64	80.50	+15.14	95.63
VK-RCD-HAR-Teacher	random_splitvk+depth+lidar+mmwave		95.57	72.20	+23.37	95.56
VK-RCD-HAR-Student-VK-Missing	random_splitvk		65.31	26.50	+38.81	65.70
VK-RCD-HAR-Student-VK-Missing	random_splitdepth		88.02	48.10	+39.92	87.86
VK-RCD-HAR-Student-VK-Missing	random_splitlidar		40.80	52.70	-11.90	40.10
VK-RCD-HAR-Student-VK-Missing	random_splitmmwave		85.88	85.70	+0.18	86.06
VK-RCD-HAR-Student-VK-Missing	random_splitvk+depth		89.51	45.30	+44.21	89.36
VK-RCD-HAR-Student-VK-Missing	random_splitvk+lidar		68.54	35.20	+33.34	69.05
VK-RCD-HAR-Student-VK-Missing	random_splitvk+mmwave		92.75	73.40	+19.35	92.82
VK-RCD-HAR-Student-VK-Missing	random_splitdepth+lidar		88.52	51.60	+36.92	88.39
VK-RCD-HAR-Student-VK-Missing	random_splitdepth+mmwave		96.45	79.80	+16.65	96.44
VK-RCD-HAR-Student-VK-Missing	random_splitlidar+mmwave		89.51	88.70	+0.81	89.57
VK-RCD-HAR-Student-VK-Missing	random_splitvk+depth+lidar		89.62	48.70	+40.92	89.49

??	??	????	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD-HAR-Student-VK-Missing	random_splitvk+depth+mmwave		96.56	70.70	+25.86	96.55
VK-RCD-HAR-Student-VK-Missing	random_splitvk+lidar+mmwave		92.73	77.80	+14.93	92.81
VK-RCD-HAR-Student-VK-Missing	random_splitdepth+lidar+mmwave		96.44	80.50	+15.94	96.43
VK-RCD-HAR-Student-VK-Missing	random_splitvk+depth+lidar+mmwave		96.56	72.20	+24.36	96.55
VK-RCD-HAR-Teacher	random_splitvk		3.40	26.50	-23.10	0.45
VK-RCD-HAR-Teacher	random_splitdepth		83.71	48.10	+35.61	82.80
VK-RCD-HAR-Teacher	random_splitlidar		3.45	52.70	-49.25	1.22
VK-RCD-HAR-Teacher	random_splitmmwave		66.72	85.70	-18.98	64.78
VK-RCD-HAR-Teacher	random_splitvk+depth		83.50	45.30	+38.20	82.60
VK-RCD-HAR-Teacher	random_splitvk+lidar		3.85	35.20	-31.35	1.26
VK-RCD-HAR-Teacher	random_splitvk+mmwave		66.46	73.40	-6.94	63.67
VK-RCD-HAR-Teacher	random_splitdepth+lidar		83.39	51.60	+31.79	82.57
VK-RCD-HAR-Teacher	random_splitdepth+mmwave		95.76	79.80	+15.96	95.74
VK-RCD-HAR-Teacher	random_splitlidar+mmwave		66.64	88.70	-22.06	63.92
VK-RCD-HAR-Teacher	random_splitvk+depth+lidar		83.70	48.70	+35.00	82.97
VK-RCD-HAR-Teacher	random_splitvk+depth+mmwave		95.66	70.70	+24.96	95.65
VK-RCD-HAR-Teacher	random_splitvk+lidar+mmwave		66.33	77.80	-11.47	63.31
VK-RCD-HAR-Teacher	random_splitdepth+lidar+mmwave		95.62	80.50	+15.12	95.61

??	??	????	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD-HAR-Teacher	random_splitvk+depth+lidar+mmwave		95.59	72.20	+23.39	95.58
VK-RCD-HAR-Ablation-NoKD	random_splitvk		60.94	26.50	+34.44	59.31
VK-RCD-HAR-Ablation-NoKD	random_splitdepth		86.85	48.10	+38.75	86.78
VK-RCD-HAR-Ablation-NoKD	random_splitlidar		33.14	52.70	-19.56	31.11
VK-RCD-HAR-Ablation-NoKD	random_splitmmwave		83.13	85.70	-2.57	83.31
VK-RCD-HAR-Ablation-NoKD	random_splitvk+depth		88.09	45.30	+42.79	87.94
VK-RCD-HAR-Ablation-NoKD	random_splitvk+lidar		63.63	35.20	+28.43	62.65
VK-RCD-HAR-Ablation-NoKD	random_splitvk+mmwave		91.05	73.40	+17.65	91.12
VK-RCD-HAR-Ablation-NoKD	random_splitdepth+lidar		87.24	51.60	+35.64	87.25
VK-RCD-HAR-Ablation-NoKD	random_splitdepth+mmwave		95.57	79.80	+15.77	95.56
VK-RCD-HAR-Ablation-NoKD	random_splitlidar+mmwave		85.92	88.70	-2.78	86.09
VK-RCD-HAR-Ablation-NoKD	random_splitvk+depth+lidar		88.62	48.70	+39.92	88.53
VK-RCD-HAR-Ablation-NoKD	random_splitvk+depth+mmwave		95.96	70.70	+25.26	95.95
VK-RCD-HAR-Ablation-NoKD	random_splitvk+lidar+mmwave		91.23	77.80	+13.43	91.29
VK-RCD-HAR-Ablation-NoKD	random_splitdepth+lidar+mmwave		95.50	80.50	+15.00	95.50
VK-RCD-HAR-Ablation-NoKD	random_splitvk+depth+lidar+mmwave		95.99	72.20	+23.79	95.98
VK-RCD-HAR-Ablation-UniformFusion	random_splitvk		5.52	26.50	-20.98	1.32

??	??	????	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD- HAR-Ablation- UniformFusion	random_splitdepth		85.59	48.10	+37.49	85.30
VK-RCD- HAR-Ablation- UniformFusion	random_splitlidar		2.65	52.70	-50.05	1.55
VK-RCD- HAR-Ablation- UniformFusion	random_splitmmwave		69.91	85.70	-15.79	68.71
VK-RCD- HAR-Ablation- UniformFusion	random_splitvk+depth		86.48	45.30	+41.18	86.24
VK-RCD- HAR-Ablation- UniformFusion	random_splitvk+lidar		4.58	35.20	-30.62	1.76
VK-RCD- HAR-Ablation- UniformFusion	random_splitvk+mmwave		67.75	73.40	-5.65	66.21
VK-RCD- HAR-Ablation- UniformFusion	random_splitdepth+lidar		86.16	51.60	+34.56	85.91
VK-RCD- HAR-Ablation- UniformFusion	random_splitdepth+mmwave		95.54	79.80	+15.74	95.54
VK-RCD- HAR-Ablation- UniformFusion	random_splitlidar+mmwave		67.80	88.70	-20.90	66.56
VK-RCD- HAR-Ablation- UniformFusion	random_splitvk+depth+lidar		86.43	48.70	+37.73	86.19
VK-RCD- HAR-Ablation- UniformFusion	random_splitvk+depth+mmwave		95.50	70.70	+24.80	95.49
VK-RCD- HAR-Ablation- UniformFusion	random_splitvk+lidar+mmwave		67.09	77.80	-10.71	65.08

??	??	????	Ours Acc.	X-Fi Acc.	Δ Acc.	Macro-F1
VK-RCD- HAR-Ablation- UniformFusion	random_splitdepth+lidar+mmwave		95.42	80.50	+14.92	95.42
VK-RCD- HAR-Ablation- UniformFusion	random_splitvk+depth+lidar+mmwave		95.39	72.20	+23.19	95.39

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?????? reports/ ???? CSV ????????????????

- reports/completion_audit.csv
- reports/HPE/hpe_main_results.csv
- reports/HPE/hpe_robustness_results.csv
- reports/HPE/hpe_official_comparison.csv
- reports/HAR/har_main_results.csv
- reports/HAR/har_official_comparison.csv
- reports/official_comparison_all.csv
- HPE/legacy_xfi/xfi_hpe_table1.csv
- HAR/legacy_xfi/xfi_har_table6.csv